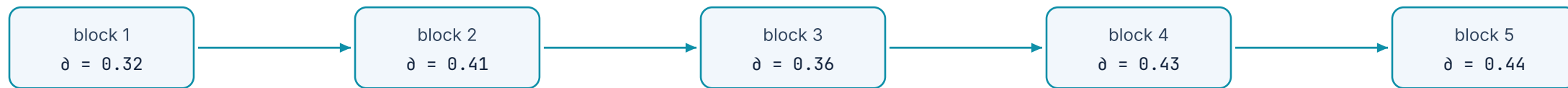
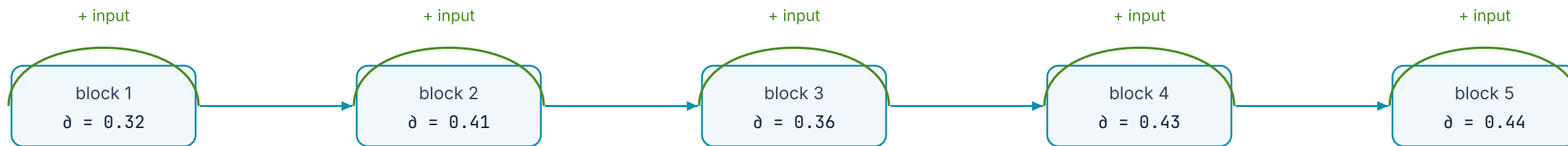


Plain stack



With residual connections



gradient reaching block 1

$$0.32 \times 0.41 \times 0.36 \times 0.43 \times 0.44 = 0.0089$$

$$(1 + 0.32) \times (1 + 0.41) \times (1 + 0.36) \times (1 + 0.43) \times (1 + 0.44) = 5.21$$

583× larger

A product of numbers below one goes to nothing. Adding the input makes every factor bigger than one, so it cannot.